

δ -Aminolevulinic Acid Synthase

(See Figs. 132-1 and 132-2)

Heme synthesis begins in the mitochondrion, where a single molecule of glycine condenses with succinyl coenzyme A (SCoA) to form δ -aminolevulinic acid (ALA), a five-carbon aminoketone. This reaction is catalyzed by the mitochondrial enzyme **δ -aminolevulinic acid synthase (ALA synthase)** and requires **pyridoxal 5'-phosphate (PLP)** as a cofactor.

ALA synthase exists as a homodimer composed of two identical catalytically active subunits, each with a molecular mass of 55 kd. These subunits may associate with higher-molecular-mass proteins that are catalytically inactive. [3,8,9]

The enzyme plays a key regulatory role in controlling the rate of heme synthesis in the liver through a **negative feedback mechanism: heme** inhibits ALA synthase at multiple levels, including repression of gene expression, inhibition of mitochondrial import, and accelerated degradation of the enzyme.

Two differentially expressed isoforms of ALA synthase have been identified: - **ALAS1 (housekeeping form)**: Ubiquitously expressed, including in the liver; subject to feedback regulation by heme. - **ALAS2 (erythroid-specific form)**: Expressed primarily in developing erythroid cells; regulated by iron availability and erythropoietic demand, rather than heme feedback.

This differential regulation allows for tissue-specific control of heme production—tight regulation in the liver to prevent accumulation of toxic intermediates, and demand-driven synthesis in erythroid precursors for hemoglobin production.

Major Steps in the Porphyrin-Heme Biosynthetic Pathway (See Fig. 132-2)

1. **δ -Aminolevulinic acid (ALA) formation:**

Glycine + succinyl-CoA $\rightarrow$ ALA (catalyzed by ALA synthase; occurs in mitochondria).

2. **Porphobilinogen (PBG) formation:**

Two molecules of ALA condense to form one molecule of PBG (catalyzed by ALA dehydratase; occurs in cytosol).

3. **Hydroxymethylbilane formation:**

Four molecules of PBG are linked by PBG deaminase to form a linear tetrapyrrole, hydroxymethylbilane (HMB; in cytosol).

4. **Uroporphyrinogen (UROGEN) III formation:**

HMB is converted to **UROGEN III** by uroporphyrinogen III synthase. This involves inversion ("flipping") of one pyrrole ring, which determines the correct side-chain arrangement for heme synthesis.

Note: In the absence of this enzyme, HMB cyclizes spontaneously to **UROGEN I**, which is not a physiological intermediate.

5. **Coproporphyrinogen (COPROGEN) III formation:**

UROGEN III undergoes sequential decarboxylation of its four acetyl groups to methyl groups, forming **COPROGEN III** (catalyzed by uroporphyrinogen decarboxylase; in cytosol).

6. **Protoporphyrinogen (PROTOGEN) IX formation:**

COPROGEN III is oxidized and decarboxylated by **coproporphyrinogen oxidase** to form **PROTOGEN IX** (occurs in mitochondria; involves oxidative decarboxylation of two propionyl side chains to vinyl groups).

7. **Protoporphyrin IX (PPIX) formation:**

PROTOGEN IX is oxidized to **protoporphyrin IX** by protoporphyrinogen oxidase (in mitochondria).

8. **Heme formation:**

Ferrous iron (Fe^{2+}) is inserted into PPIX by **ferrochelatase** (also in mitochondria), forming **heme**.

Heme Catabolism

After degradation of hemoproteins (e.g., hemoglobin), heme is metabolized via:

- **Heme oxygenase**: Cleaves heme to produce **biliverdin**, releasing iron and carbon monoxide. - **Biliverdin reductase**: Converts biliverdin to **bilirubin** (a linear tetrapyrrole). - Further metabolism leads to conjugation and excretion as bile pigments.

This catabolic pathway is catalyzed by microsomal enzymes, including **NADPH-cytochrome P450 reductase**, **heme oxygenase**, and **biliverdin reductase**.

Diagnosis of Porphyria

Definitive diagnosis requires one or more of the following: 1. Measurement of **porphyrins** and/or **porphyrin precursors** (e.g., ALA, PBG) in urine, stool, plasma, or erythrocytes. 2. Assay of **residual activity** of specific heme biosynthetic enzymes. 3. Identification of **pathogenic mutations** in genes encoding these enzymes.

Clinical suspicion is often raised by characteristic symptoms and family history, but laboratory confirmation is essential.

Modern molecular techniques have clarified the genetic and biochemical basis of the **porphyrias**, a group of disorders caused by defects in heme biosynthesis, leading to accumulation of pathway intermediates.

The biosynthetic steps and regulatory mechanisms are summarized in Figs. 132-1 through 132-4.